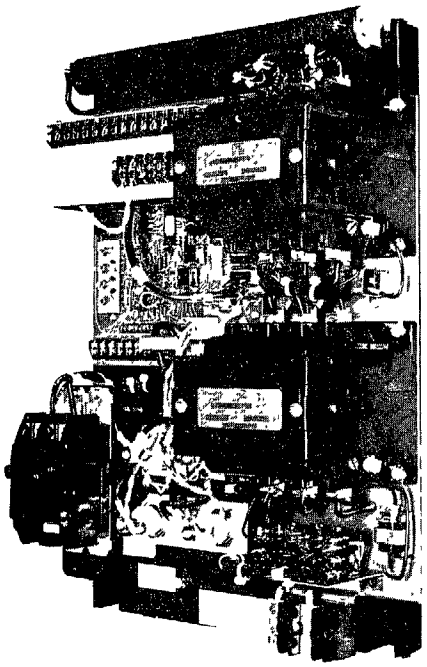
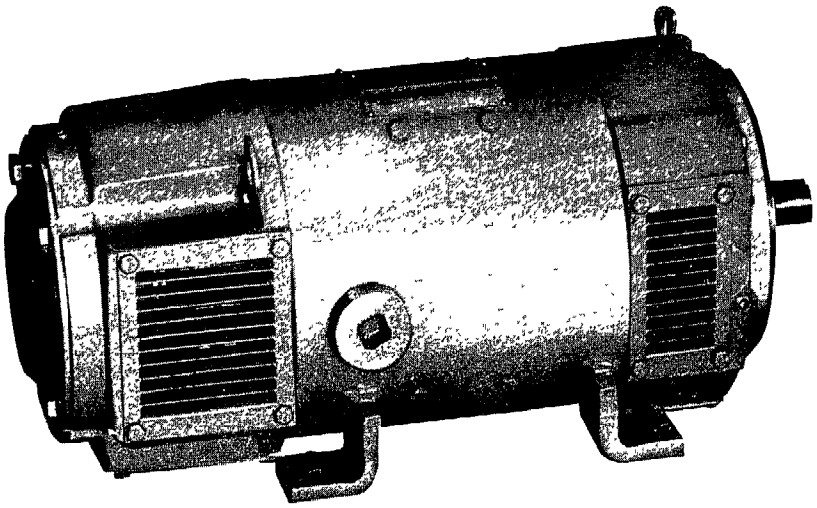
STATOTROL* II ADJUSTABLE SPEED DRIVE 6VFWC . . . . . SERIES

7½ HP SINGLE PHASE

230 VOLTS, 50/60 HERTZ





DEPENDABLE MOTOR CONTROL

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company

GENERAL ELECTRIC

*Registered Trademark of the General Electric Co., U.S.A.

TABLE OF CONTENTS

Page

INTRODUCTION 3
DESCRIPTION 3
RECEIVING AND STORAGE 3
INSTALLATION 3
ELECTRICAL CONNECTIONS 3
PREPOWER CHECKS AND ADJUSTMENTS 6
INITIAL OPERATION AND ADJUSTMENT 7
OPERATOR CONTROLS, NORMAL OPERATION 8
MAINTENANCE 9
TROUBLESHOOTING AND REPAIR 9
RENEWAL PARTS 13
WARRANTY 15

LIST OF ILLUSTRATIONS

FIGURE

1. Outline and Mounting Dimensions Statotrol II 7.5 HP Controller 4
2. Power Wiring Connection Diagram 4
3A. Typical Remote Control Station Connection Diagram 5
3B. Remote Adapter Connections 5
4. Remote Accessory Connection Diagram 6
5. 50 Hertz Adjustments 7
6. Elementary Diagram Statotrol II DC Drive 7.5 HP Controller 14

LIST OF TABLES

TABLE

I. Minimum Wire and Circuit Breaker Size 6
II. Prepower Wiring Checks 7
III. Troubleshooting Guide . . . . . 10 - 11 - 12

STATOTROL II 7.5 HP DC DRIVE

INTRODUCTION

This manual contains general information on the General Electric Statotrol II DC drive. Additional information and diagrams may be supplied with the equipment when necessary.

DESCRIPTION

The Statotrol II drive is a high performance full wave DC drive. The regulating function is provided by a custom integrated circuit. The motor is a shunt wound field DC motor specially designed for use with a full wave phase controlled SCR power supply.

Standard features of the Statotrol II drive include speed control, protective current limit, undervoltage protection, and potentiometer adjustments for positive and negative IR compensation, maximum speed, minimum speed, and torque taper. Optional features include dynamic braking, reversing, adjustable torque control, independently adjustable linear timed acceleration and deceleration, tach feedback, instrument follower, tach follower, speed indication, regenerative braking, and a variety of special functions.

RECEIVING AND STORAGE

As soon as the equipment is received, it should be unpacked and examined for any damage sustained in transit. If damage is evident, a damage claim should be filed immediately with the transportation company, and the local General Electric Company Sales Office should be promptly notified.

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected against accidental damage. Avoid storage in a location where construction work is in progress.

INSTALLATION

CONTROLLER INSTALLATION

The Statotrol II controller must be mounted in a location which will allow free flow of cooling air over the fins on the heat sink. Maximum ambient temperature around the controller must not exceed 40°C (104°F). Figure 1 gives mounting dimensions and outline dimensions. It is recommended that two or more inches of clearance be provided all around the controller to assure adequate air flow through the heat sink fins.

MOTOR INSTALLATION

The motor must be firmly mounted and properly aligned to prevent vibration. Excessive vibration causes rapid wear and objectionable audio noise. Heat dissipated by the motor will raise the ambient temperature around the motor if the motor is installed in an enclosed space. Since a lower ambient temperature will extend the motor life, it is always desirable to mount the motor in a well ventilated location. The motor should never be used where the ambient temperature exceeds 40°C (104°F) unless an oversized motor has been specially selected. Refer to the instruction book provided with the motor for specific installation instructions.

WARNING

IF THE MOTOR IS ACCESSIBLE WHILE IT IS RUNNING, A PROTECTIVE GUARD SHOULD BE INSTALLED AROUND ALL EXPOSED ROTATING PARTS.

ELECTRICAL CONNECTIONS

PROPER LINE VOLTAGE CONNECTION

This controller operates on 50/60 cycle 230 volt AC. A transformer is required where supply voltage deviates more than 10% from the controller rating. A calibration resistor 140R must be removed from the main component board if 50 Hz power is to be used. Refer to Figure 5.

GROUNDING

WARNING

PERSONNEL SAFETY CONSIDERATIONS AND THE NATIONAL ELECTRIC CODE REQUIRE THAT ELECTRICAL APPARATUS ENCLOSURES BE SOLIDLY CONNECTED TO BUILDING GROUND.

The necessary wire gauge of the ground wire is determined by the rating of the branch circuit breaker.

Connect a suitably grounded lead to the ground connection terminal provided in the Statotrol II enclosure near the circuit breaker.

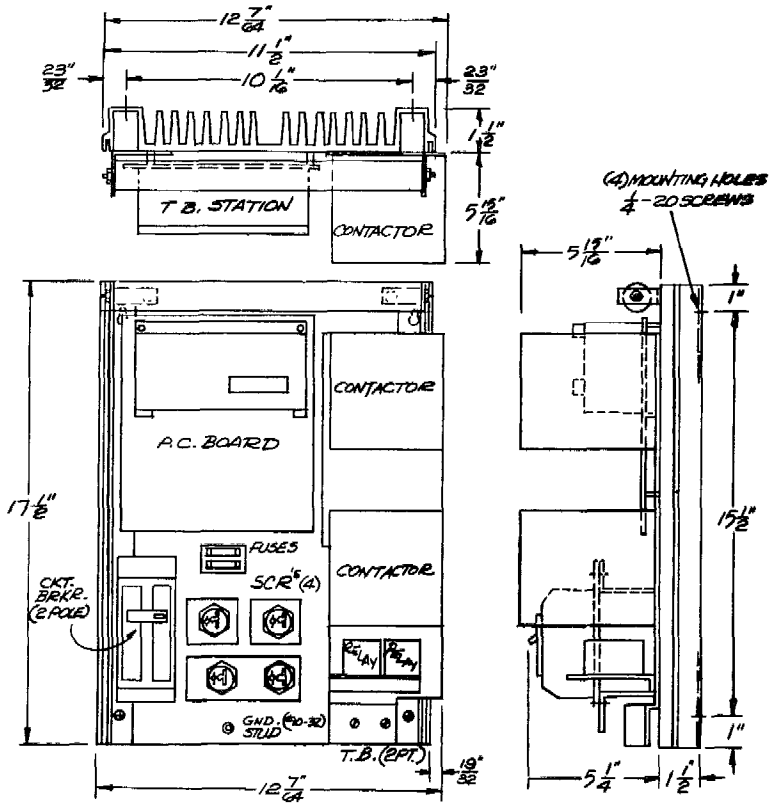


Figure 1. Statotrol II Controller, 7.5 HP, Outline and Mounting Dimensions

WARNING

PROPER MOTOR GROUNDING IS ESSENTIAL FOR PERSONNEL SAFETY. DO NOT DEPEND UPON MOTOR MOUNTING BOLTS TO GROUND THE MOTOR FRAME. A 1/4-20 BOLT IN A TAPPED HOLE IN THE MOTOR END SHIELD IS PROVIDED FOR THE GROUND CONNECTION.

POWER WIRING

Connect 230V 50 or 60 Hz lines to the controller circuit breaker. The AC power must not vary from nominal voltage by more than 10%. Resistor 140R (see Figure 5) must be cut out of the circuit before 50 Hz power is connected. Table I gives minimum required circuit breaker and wire sizes.

Figure 2 is a wiring diagram for the power connections. When the motor is connected as shown in Figure 2, the motor will rotate CCW as viewed from the opposite shaft end. Turn off the circuit breaker and reverse leads A1 and A2 for clockwise rotation.

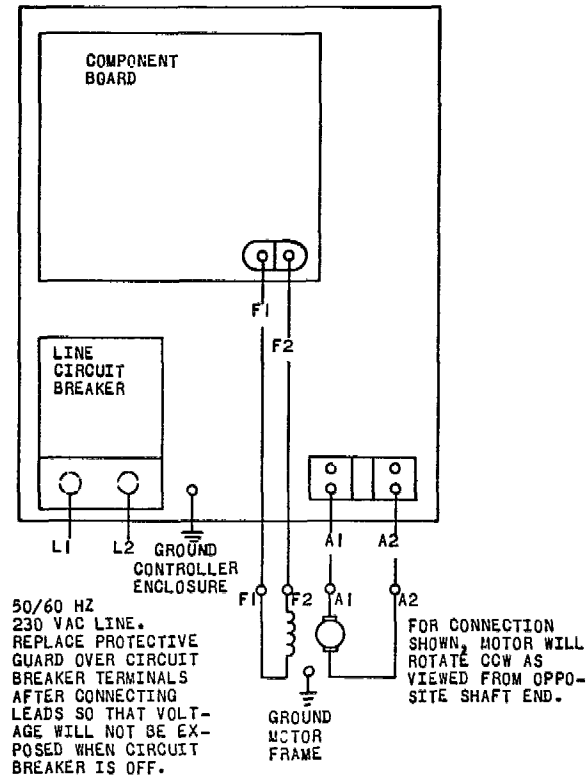


Figure 2. Power Wiring Connection Diagram

CAUTION

ANY CONNECTION ERROR IN THE POWER WIRING CAN DAMAGE THE CONTROL. RECHECK EACH CONNECTION BEFORE POWER IS APPLIED.

If there is any doubt about motor lead identification, the problem can be resolved with a simple resistance check. The motor armature resistance will be about 0.3 ohms to 1 ohm, while field resistance will be about 200 ohms to 500 ohms. There must be no continuity between armature and field.

Refer to the instruction book packed with the motor for specific motor connection instructions. If the motor has leads marked C1 and C2, these leads should be independently taped and well insulated and left unconnected.

The branch circuit breaker must be large enough to eliminate nuisance tripping and small enough to protect the branch circuit and motor connection conductors. Refer to the National Electric Code (**1975**, section 310-20) and Table I on Page 6 for minimum circuit breaker and wire sizes. Wherever local codes are more restrictive, the local codes take precedence over the National Electric Code. Special requirements exist for installations in hazardous locations and other special situations.

A line power isolation transformer must be used when the Statotrol drive is connected to an instrument signal source which is not isolated from ground. When a line power transformer is used, any line switch used to disconnect the controller should be on the load side of the transformer.

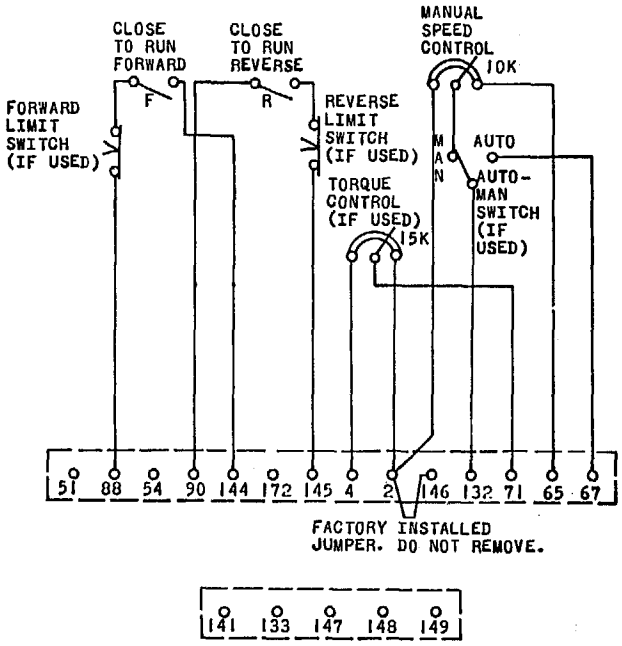
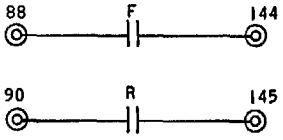
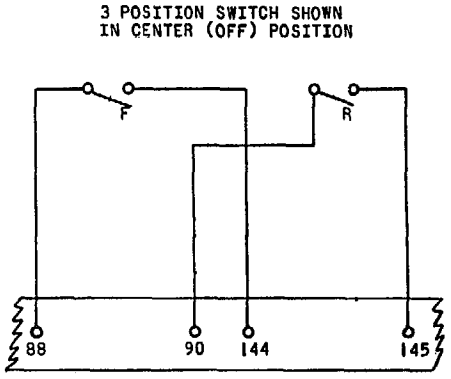


Figure 3A. Typical Remote Control Station Connection Diagram



Connections for Forward, Reverse, Stop Switching by Pair of 2 Pole Relays



NOTE: Close F to run forward, close R to run reverse.

Partial Connection Diagram Showing Connection for 4 Pole 3 Position Forward, Reverse, Off Switch

Figure 3B. Remote Adapter Connections

REMOTE CONTROL STATION WIRING

When a remote control station is to be installed, it must be connected to the 6VRC2002 series adapter unit which plugs into the Statotrol II controller main component board.

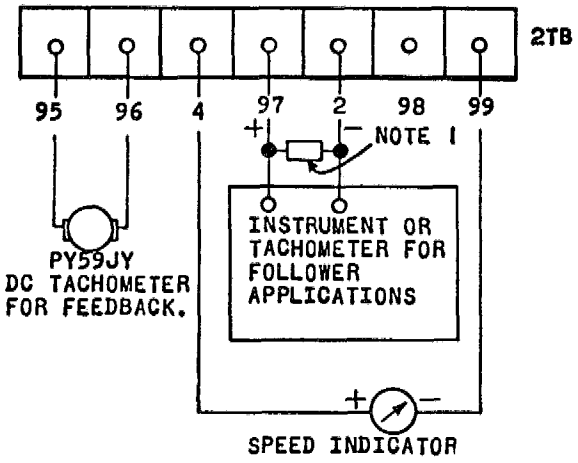
Conductors connected to the speed adjustment potentiometer meet the Class I control and signal circuit requirements of Article 725 of the National Electric Code (1975). Wherever local codes are more restrictive, the local codes take precedence over the National Electric Code.

Figure 3B shows various remote control station connec-

TABLE I

STANDARD DRIVES HORSEPOWER RATING	MINIMUM WIRE AND CIRCUIT-BREAKER SIZE
7.5 Horsepower	60 AMP Circuit Breaker 75°C #4 AWG copper wire for line and armature 75°C #14 AWG copper wire for motor field

tion schemes which can be used with the 6VRC2002A1 remote adapter unit.



NOTE 1:

Protective shunt resistor is required to prevent overvoltage when a current signal instrument with maximum output of more than 5 milliamps is used. Use a 560 ohm resistor with a 20 ma max. signal, and use a 180 ohm resistor with a 50 ma max. signal. The proper value of the protective shunt resistor can be found by the following formula when instruments with other maximum outputs are used.

$$R = \left(\frac{8500}{I - 5}\right) \text{ ohms } \pm 10\%$$

R is the resistance of the protective shunt resistor and I is the maximum output of the signal generator in milliamps.

NOTE 2:

See instruction sheet provided with option for connection diagram when instrument with voltage signal output is used in follower applications.

Figure 4. Remote Accessory Connection Diagram

CAUTION

TO AVOID ELECTRICAL NOISE PICKUP, IT IS NECESSARY TO KEEP THE TORQUE POTENTIOMETER, SPEED POTENTIOMETER AND AUTO-MANUAL SWITCH WIRES SEPARATE FROM ALL OTHER WIRES. DO NOT RUN THESE WIRES THROUGH CONDUITS WITH POWER CONDUCTORS OR RELAY COIL WIRING.

MOTOR THERMOSTAT CONNECTIONS

When a motor thermostat is used, it should be connected in series with the "off" switch. Refer to Figure 3A for connection diagrams of remote control stations with motor thermostats.

REMOTE ACCESSORY WIRING

A seven point terminal strip, 2TB, is provided for connection of leads from remote accessories. This terminal strip is not used unless one of the optional remote accessory adapter units has been installed in the Statotrol controller. Remote accessory functions include tachometer feedback, tachometer follower, instrument follower, and speed indication.

CAUTION

ALL CONNECTIONS TO REMOTE ACCESSORY OPTION MUST BE PHYSICALLY SEPARATED FROM ALL OTHER WIRES. IF THIS IS NOT FEASIBLE, SHIELDED WIRE MUST BE USED. CONNECT SHIELD LEAD TO 2TB(2). DO NOT GROUND SHIELD. SHIELD MUST BE INSULATED TO PREVENT ACCIDENTAL CONTACT TO BUILDING GROUND.

Figure 4 shows the wiring connections for remote accessories. The wiring shown should be considered as Class I signal wiring (National Electric Code, 1975, Article 725) except the instrument signal wiring which may be considered Class II if the instrument output is limited as specified in the referenced code article.

PREPOWER CHECKS AND ADJUSTMENTS

WARNING

THIS SECTION CONTAINS IMPORTANT WARNINGS AND CAUTIONS WHICH MUST BE OBSERVED DURING INSTALLATION OF THE CONTROL AND MOTOR. FAILURE TO OBSERVE THESE CAUTIONS MAY CAUSE SAFETY HAZARDS OR EQUIPMENT DAMAGE. READ THIS SECTION CAREFULLY AND MAKE ALL NECESSARY ADJUSTMENTS BEFORE POWER IS APPLIED TO THE CONTROL.

CURRENT LIMIT SETTING

The current limit circuit is factory adjusted for the motor supplied. If another motor is used or if the current limit setting appears to be wrong, contact the factory for instructions.

50 CYCLE POWER ADJUSTMENT

The controller is factory adjusted for use with 60 Hz AC power. If 50 Hz power is to be used, resistor 140R must be

cut out of the circuit before 50 Hz power is applied. See Figure 5 for the location of 140R.

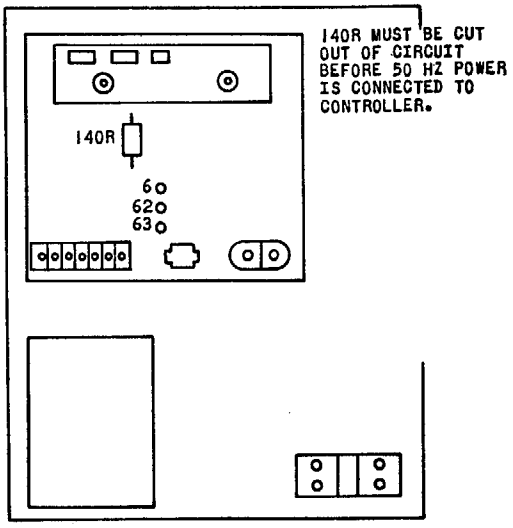


Figure 5. 50 Hz Adjustment

MOTOR CHECKS

If the motor is accessible while it is running, a protective guard should be installed around all exposed rotating parts.

The "Inspection Before Starting" section of the motor instruction book must be carefully followed.

WIRING CHECK

Any connection error in the power wiring can damage the control as soon as power is applied. Any short to building ground in the signal wiring can cause damage as soon as power is applied, unless the Statotrol II drive is connected to a line power isolation transformer. If the controller is connected to any instruments or equipment which may be grounded, a line isolation transformer must be used. In no case should the circuit be grounded at more than one point. If the drive is to be operated as an instrument follower, verify that the correct protective resistor is properly installed. (Refer to Figure 4.) It is advisable to recheck all wiring before power is applied. Make certain that all screw terminals are tight. If an ohmmeter is available, the simple resistance checks shown in Table II should be performed to verify that the power wiring is correct.

The relay coil circuit is normally grounded at circuit point 51, and is isolated from the other controller circuits.

GROUNDING

Safety considerations and the National Electric Code require the motor frame and the controller enclosure to be connected solidly to building ground. Do not rely on mounting bolts for grounding. Refer to the "Electrical Connections" section of this instruction book for ground lead connections.

TABLE II
PREPOWER WIRING CHECKS

TERMINALS	APPROPRIATE RESISTANCE
Load side of open circuit breaker	**300 to 1000 ohms**
F1 to F2	200 to 500 ohms
A1 to A2	0.3 to 1.0 ohms
F1 to F2 or A1 to A2	500,000 ohms or more
Control Circuit to Ground	500,000 ohms or more (see exceptions in preceding paragraphs).

INITIAL OPERATION AND ADJUSTMENT

When the control has been mounted and connected, and the prepower checks and adjustments are complete, apply power to the control, turn the speed control knob to zero, and press the "Start" button on the control station. Slowly turn the speed control knob until the motor starts to turn. Check the direction of motor rotation to be sure it is correct. If the motor turns the wrong way, disconnect power from the control and reverse the motor armature leads.

The "Inspection After Starting" section of the motor instruction book should be carefully followed at this time.

The control is now ready for normal operation. Some Applications may require special settings for maximum speed, minimum speed, IR compensation, and torque taper. Potentiometers for adjusting these functions are in a line along one edge of the main component board inside the controller. Remove the controller cover and use a small screwdriver to adjust the potentiometers as described in the following instructions.

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. LINE

VOLTAGE IS EXPOSED IF THE LINE CIRCUIT BREAKER IS ON AND THE COVER IS REMOVED. ALWAYS TURN OFF THE CIRCUIT BREAKER BEFORE PERFORMING ANY WORK OTHER THAN POTENTIOMETER ADJUSTMENTS INSIDE THE CONTROLLER. INTERNAL ADJUSTMENTS SHOULD BE MADE BY QUALIFIED ELECTRICIANS AND NOT BY MACHINE OPERATORS.

MINIMUM SPEED ADJUSTMENT (4P)

The minimum speed adjustment is factory set so that the motor will start to run when the speed control knob is turned just slightly off zero. To change this setting, first turn the speed control knob to zero and then rotate the minimum speed potentiometer until the motor runs at the desired minimum speed.

The minimum speed adjustment is also used to match proper motor speed to specific signal levels when the reference signal comes from a signal generating instrument. This function is explained on the instruction sheets for the individual remote accessory adapter option boards.

Adjusting the minimum speed changes the maximum speed also, so the minimum speed should be set before the maximum speed potentiometer is adjusted.

MAXIMUM SPEED ADJUSTMENT (3P)

CAUTION

DO NOT CHANGE THE FACTORY SETTING OF THE MAXIMUM SPEED POTENTIOMETER IF TACHOMETER FEEDBACK IS TO BE USED. IF THE MAXIMUM SPEED POTENTIOMETER IS INADVERTENLY DISTURBED, IT SHOULD BE RETURNED TO APPROXIMATELY 80% OF ITS POSSIBLE CW ROTATION.

THE MAXIMUM SPEED ADJUSTMENT DOES NOT LIMIT MOTOR SPEED WHEN THE CONTROL IS IN THE "AUTOMATIC" MODE OF OPERATION (TACHOMETER FOLLOWER AND INSTRUMENT FOLLOWER APPLICATIONS).

The maximum speed adjustment is factory set to a speed slightly higher than rated motor speed. To change the maximum speed, first turn the speed control knob to its highest speed setting. Then adjust the maximum speed potentiometer until the motor runs at the highest speed needed for the particular application.

The motor should never be operated above its rated speed.

IR COMPENSATION ADJUSTMENT (6P)

Conventional IR compensation boosts the motor speed at high load. The Statotrol II IR compensation circuit provides this feature and can also be adjusted to provide "droop" for applications which require motor speed to drop off linearly as load is added. The IR compensation adjustment factory setting is adequate for most applications. When necessary, the IR compensation can be adjusted by the following procedure. First set the speed control knob to the speed at which regulation is most critical and measure the motor speed precisely with a tachometer or a strobe light. Load the motor (motor may be turned off while load is added) and then turn the IR compensation potentiometer until the motor runs at the desired speed. There is no need to wait for the motor to warm up since the Statotrol II IR compensation is relatively insensitive to motor temperature. Some loads may "hunt" if the IR compensation is set too high. Turning the IR compensation potentiometer CCW should eliminate this hunting.

TORQUE TAPER ADJUSTMENT (5P)

CAUTION

THE TORQUE TAPER ADJUSTMENT IS FACTORY SET FULL CCW. THE TORQUE TAPER ADJUSTMENT MUST BE KEPT FULL CCW WHEN A CONTROL STATION WITH A TORQUE CONTROL KNOB IS USED.

In applications which require a fixed program of torque control, the torque taper adjustments may be turned clockwise to increase the output torque which the motor will deliver at low speeds.

OPERATOR CONTROLS, NORMAL OPERATION

The operator controls have been made as simple and foolproof as possible. However, in certain applications, it may be necessary to caution the machine operator against operation sequences which may damage the machinery or process driven by the Statotrol motor. The following instructions apply only when the customer's load requires no special operating sequences.

STARTING

Press the "start", "forward", or "reverse" button. The

speed control knob may be at any desired setting. The motor will accelerate smoothly to the speed set by the speed control knob.

Motor current is automatically limited to a safe value. No warmup is required, and motor response is immediate.

CHANGING SPEED

The speed control knob may be turned to any desired setting at any time, and the motor will respond smoothly. Turning the speed control knob clockwise will increase the motor speed.

STOPPING

Press the "stop" button. Motor will stop smoothly regardless of setting of knobs and other switches.

TORQUE CONTROL

The torque control knob adjusts the maximum output torque the motor will deliver. To operate in the torque control mode, first set the speed control knob to the highest desired speed and then turn the torque control knob to any desired setting.

Torque knob settings above 5 on the dial are for intermittent duty only, since settings above 5 may cause the motor to deliver more than rated torque (except with limited torque loads) and the motor will consequently overheat. The numbers on the torque control knob are for indication only, and should not be relied upon for motor overload protection. When the motor load is such that it will not overload the motor except during acceleration, the torque control knob may be left at any setting if the motor is not stopped and restarted continuously.

REVERSING

The rotation of the motor may be reversed by pressing the stop button and allowing motor to come to a complete stop, then pressing the "reverse" button.

MAINTENANCE

CONTROLLER

The controller enclosure should be periodically inspected to prevent an accumulation of materials which might block the flow of cooling air through the heat sink fins. The control must be protected from dripping or sprayed moisture and oil.

MOTOR

Complete instructions for motor maintenance are included in the motor instruction book.

GEAR BOX

Periodic oil changes and bearing lubrications are necessary to prolong the life of most gear boxes. Refer to the instructions provided with the unit for recommended maintenance schedule and lubricants.

TACHOMETER

After 5 years of normal service or 2 years of severe duty, the bearings should be inspected. If they appear to be loose or worn, they should be replaced. Brushes should be inspected after every 1000 hours of operation and replaced when they wear down to 3/8" in length.

TROUBLESHOOTING AND REPAIR

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. LINE VOLTAGE IS EXPOSED IF THE LINE CIRCUIT BREAKER IS ON WHILE THE CONTROLLER COVER IS REMOVED. USE EXTREME CARE TO AVOID TOUCHING EXPOSED CONDUCTORS. ALWAYS TURN OFF THE LINE CIRCUIT BREAKER BEFORE DOING ANYTHING OTHER THAN ADJUSTING POTENTIOMETERS. THE CONTROL STATION STOP BUTTON DOES NOT REMOVE POWER FROM THE CONTROL OR THE MOTOR FIELD.

If a newly installed drive will not function properly, it is most likely that a terminal is loose, or a problem exists with a connection, line voltage, or an adjustment. Line voltage must be between 253 volts AC and 207 volts AC, and the adjustments must be set as described in the "Initial Operation and Adjustments" section of this book.

If the drive operates normally for a while and then malfunctions, the problem may be line voltage, motor overload, motor failure, a loose terminal, an open fuse, or a component failure. In the following discussions, each step of troubleshooting is based on the assumption that all preceding steps have been completed and nothing abnormal has been found.

SYMPTOM	PRIMARY CAUSES	SECONDARY CAUSES	SOLUTIONS
Motor will not run when Run button is pushed.	No AC voltage being applied to controller.	Branch circuit breaker or controller circuit breaker tripped or defective.	Reset circuit breaker or replace if defective.
	No field voltage.	Fuses 5FU-6FU blown.	Replace fuses.
		Field diodes (16D-19D) defective.	Replace printed wire board.
		Field circuit foil on printed wire board open.	Replace printed wire board.
	No control voltage applied to printed wire board.	Fuses 5FU-6FU blown.	Replace fuses.
		Electrical contact lost at terminal posts on printed wire board. (Located in lower right hand corner of printed wire board, circuits 122 or 123.)	Reconnect lead if loose. If not loose, then partially extract leads from both circuit 122 and 123 terminals, then reseat to restore electrical contact.
	Motor field circuit open (external to controller).	Field circuit wires and/or connections open.	Repair or replace wires and/or connections.
		Motor field winding open.	Replace motor. NOTE: check brushes first.
	Motor armature circuit open (external to controller).	Armature circuit wires and/or connections open.	Repair or replace wires and/or connections.
		Motor armature winding open.	Replace motor.
	Control relay (CR) not picking up.	Motor thermostat circuit open.	Troubleshoot motor thermostat circuit and repair or replace defective part(s).
		External forward (Run) contact (connected between circuits 88 and 144) not closing properly.	Troubleshoot forward contact circuit and repair or replace defective part(s).

SYMPTOM	PRIMARY CAUSES	SECONDARY CAUSES	SOLUTIONS
Motor will not run when Run button is pushed (Cont.)	Control relay (CR) not picking up (Cont.)	Control relay (CR) coil voltage low or zero. (This can be measured on the printed wire board at the connector marked "M CONN." Place positive probe on female terminal and negative probe on middle right hand terminal. Should be 18-24 volts D.C.)	Check fuses 5FU and 6FU first; if they check good, replace the printed wire board.
		Diode 33D defective (located on remote control station adaptor printed wire board).	Replace remote adaptor unit.
		Electrical contact lost between remote adaptor receptacles and main printed wire board pins.	Loosen remote adaptor mounting screws and pull up and reseat it several times to restore electrical contact.
		Resistor 96R or Transistor 96Q on main printed wire board defective.	Replace printed wire board.
	Forward auxiliary relay (Faux.) not picking up.	Defective relay.	Replace relay.
		Diode 32D defective (located on remote control station adaptor printed wire board).	Replace remote adaptor unit.
		Electrical contact lost between remote adaptor receptacles and main printed wire board.	Loosen remote adaptor mounting screws and pull up and reseat several times to restore electrical contact.
		Wiring and/or wiring terminations in Faux. coil circuit defective.	Check all wires and terminations in Faux. coil circuit and repair or replace as necessary.
		Diode 25D on main printed wire board defective.	Replace printed wire board.

SYMPTOM	PRIMARY CAUSES	SECONDARY CAUSES	SOLUTIONS
Motor will not run when Run button is pushed (Cont.)	Forward Contactor (F) not picking up.	Fuses 5FU-6FU blown.	Replace fuses.
		F contactor defective.	Replace contactor.
		Suppression network 1RC defective.	Replace 1RC.
		Faux. series contact not closing.	Replace Forward Auxiliary Relay.
		Wiring and/or wiring terminations in F contactor coil circuit defective.	Check all wires and terminations in F contactor coil circuit and repair or replace as necessary.
	No firing voltage being applied to the SCR's.	Electrical contact lost between SCR's and main printed wire board.	Partially extract and then reseat each of the SCR connectors at the printed wire board to restore electrical contact.
		Main printed wire board defective.	Replace printed wire board.
Motor will not run when reverse button is pushed.	NOTE All of the primary and secondary causes and solutions listed above apply to problems encountered in reverse operation also The following substitutions should be made. Substitute Raux. for Faux. Substitute R contactor for F contactor Substitute 2RC for 1RC. Substitute Rev. contact for Forward contact. Substitute 34D for 33D. Substitute 95Q for 96Q. Substitute 35D for 32D. Substitute 26D for 25D.		
Motor runs very fast for speed setting, but very little torque is produced	Open armature feedback circuit.	Jumper 14J (located on remote accessories adaptor printed wire board) open.	Replace remote accessories adaptor printed wire board.
		Electrical contact lost between remote accessories adaptor board pins and receptacles on main printed wire board.	Loosen the remote accessories adaptor board mounting screws and pull up and reseat the board pins several times to restore electrical contact.

SYMPTOM	PRIMARY CAUSES	SECONDARY CAUSES	SOLUTIONS
Motor runs very fast for speed setting, but very little torque is produced (Cont.).	Open armature feedback circuit (cont.)	Defective main printed wire board	Replace main printed wire board.
	Low Field Voltage	Field Diodes (16D-19D) defective.	Replace main printed wire board.
Motor operates normally at no load, but will not deliver adequate torque to drive load.	Firing voltage not being applied to one or more SCR's	Electrical contact lost between SCR's and main printed wire board.	Partially extract and then reseat each of the SCR connectors at the printed wire board to restore electrical contact.
		Main printed wire board defective	Replace main printed wire board.
	Half-wave voltage being applied to motor armature.	One or more defective SCR's.	Replace defective SCR's.
	Current limit jumper setting incorrect for motor in use.	Current limit jumper missing or connected to wrong terminals.	The current limit jumper is adjusted at the factory for the motor supplied. If jumper is missing or appears to be causing a problem, contact the factory for instructions.
Circuit breaker trips when power is applied.	One or more shorted SCR's.		Replace defective SCR's,
Circuit breaker trips when Run (or Rev.) button is pushed.	Motor starting current too high.	Current limit jumper missing or connected to wrong terminals.	The current limit jumper is adjusted at the factory for the motor supplied. If missing or if it appears to be causing a problem, contact the factory for instructions.
		Motor being operated on 50 hz power with resistor 140R still in circuit.	Cut resistor 140R out of circuit (see Figure 5 for location).
	Short in motor armature circuit.	Shorted armature winding or short in armature circuit wiring.	Replace motor or wiring as appropriate.

TROUBLESHOOTING GUIDE (CONTINUED)

SYMPTOM	PRIMARY CAUSES	SECONDARY CAUSES	SOLUTIONS
Circuit breaker trips after a few minutes or a few hours of steady operation.	Motor armature current too high.	Motor overloaded.	Measure the D.C. current in the motor armature. If it exceeds the nameplate rating, the motor is running at overload. Locate cause of overload and remove.
Fuses (5FU-6FU) blow repeatedly.	Short circuit exists on load side of fuses.	Short across motor field terminals.	Locate short and remove.
		Shorted varistor 1BD.	Replace printed wire board.
		Shorted capacitor 13C.	Replace printed wire board.
		Shorted transformer 1T.	Replace printed wire board.
		Shorted coil in F contactor.	Replace F contactor.
		Shorted coil in R contactor.	Replace R contactor.
		Shorted suppression network 1RC.	Replace 1RC.
		Shorted suppression network 2RC.	Replace 2RC.
		Short in wiring or terminal connections.	Check all associated wiring and terminals and repair or replace as necessary.
		Shorted field diodes (16D-19D).	Replace printed wire board.
		Shorted field winding in motor.	Replace motor.
Motor hunts.		IR compensation potentiometer misadjusted.	Turn IR compensation potentiometer CCW until hunting stops.

TROUBLESHOOTING GUIDE (CONTINUED)

CHECKING DIODES AND SCR'S

Silicon Diodes

The characteristic of a silicon diode to block current flow in one direction and pass current freely in the other direction is used in a simple ohmmeter check as follows.

Connect ohmmeter leads across the silicon diode to be checked. (High resistance scale should be used, at least "X1000".) When the meter leads are reversed, the indicated resistance should change from infinite to some very low value. (The value will vary with different instruments.)

NOTE

Silicon diodes will usually fail either to a short or an open condition which will be quickly discovered with the above check.

Silicon Controlled Rectifiers (SCR's)

SCR's can fail in several different modes. These failures could either be a short, an open or simply a change in characteristics. For the latter two cases, a skilled technician and special test equipment would be required to determine if the SCR has failed. However, if the SCR has failed due to a short, this could be discovered with an ohmmeter. Do not overlook this one simple SCR check as follows:

Connect the leads across the SCR from anode to cathode (either polarity) with the meter on a high resistance scale, at least "X1000". Now reverse the meter leads. The indicated resistance in both directions should be infinite.

REPAIR

Motor Repair

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. THE CONTROLLER STOP SWITCH DOES NOT REMOVE VOLTAGE FROM THE MOTOR FIELD. ALWAYS TURN OFF THE LINE CIRCUIT BREAKER BEFORE ATTEMPTING TO SERVICE MOTOR.

The motor can be repaired just as a standard DC motor by any competent motor repair service. For replacement parts or motor service, take the motor model number from the motor nameplate and contact the nearest service shop authorized by the motor manufacturer.

Controller Repairs

Normal field repair should be limited to replacing component boards and options. The complexity of the test sequence required to verify proper operation of a component board after repair makes it inadvisable to attempt field repair of the printed circuit board. Replacement of the complete printed circuit board is the quickest and easiest approach to troubleshooting and repair.

When returning component boards for exchange, pack them carefully to prevent additional damage from occurring in transit.

RENEWAL PARTS

Replacement parts can be ordered from the nearest sales office of General Electric Company. Replacements for control stations and options should be ordered by the "6V" catalog number which appears on the control station or option originally purchased with the control.

Motor parts can be obtained from the nearest service shop authorized by the motor manufacturer.

PRINCIPAL RENEWAL PARTS LIST
7.5 HP STATOTROL CONTROLLER

DESCRIPTION OF PART OR ASSEMBLY	CATALOG NUMBER	CIRCUIT DESIGNATION	QTY. USED PER PANEL
Main Printed Circuit Board	**44B337316–G115**		1
Silicon Controlled Rectifiers	104X125DA113	1SCR - 4SCR	4
Fuses	104X109AD035	5FU, 6FU	2
Circuit Breaker	104X100FA003	1CB	1
Relay	104X131AB023	Faux, Raux	2
Suppression Network	104X122CA002	1RC, 2RC	2

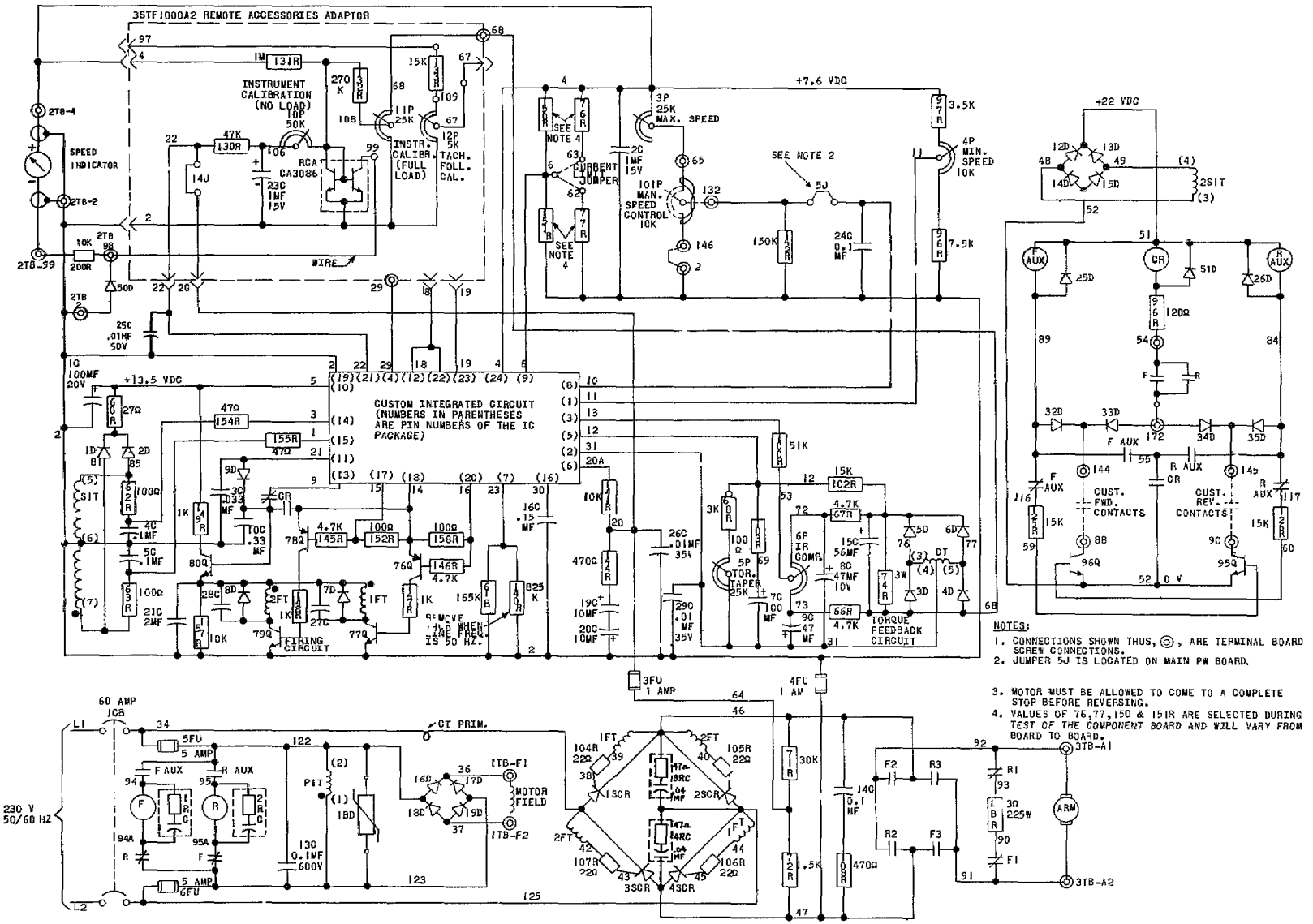


Figure 6. Statotrol II DC Drive, 7.5 HP, Elementary Diagram

WARRANTY

The Contract Warranty for Statotrol drives is stated in the General Electric Apparatus Handbook, Section 98, page 11. The purpose of the following is to provide specific instructions to the Statotrol user regarding warranty administration and how to obtain assistance on out-of-warranty failures.

STATROL II CONTROLLERS (7½ HP)

The warranty covers all major parts of the controller such as the main printed circuit board, modification kits, contactors, etc., but does not provide for replacement of fuses or the complete controller.

1. In the event of failure or misapplication during "in warranty", refer to the instruction book to identify the defective part or subassembly.

2. When the defective part has been identified (or for assistance in identification) call:

   STATOTROL DRIVE SERVICE
   Erie, Pennsylvania
   (814) 455-0521

   Before calling, list catalog numbers of the controller, motor, operator's station and any optional modifications. Include orange colored four digit identification number located in lower left corner of controller chassis.

3. Statotrol Service will ship prepaid, at no charge, any parts that have failed in warranty and advise if failed part(s) should be returned to the factory. The factory address is:

   General Electric Company
   Speed Variator Products Section, DCM&GPD
   1100 Lawrence Parkway
   Erie, Pennsylvania 16531
   Attention: STATOTROL DRIVES SERVICE

4. For parts, contact your local General Electric Sales Office or your "Authorized" General Electric Parts Distributor.

   "Exchange" parts at "Exchange" prices may be available for selected items.

STATOTROL MOTORS

Statotrol motor repairs are generally handled by General Electric Small Motor Service Stations or Apparatus Service Shops. For specific instructions on your motor, call STATOTROL DRIVE SERVICE at the number listed above and furnish complete nameplate data.

GENERAL ELECTRIC COMPANY – DIRECT CURRENT MOTOR & GENERATOR PRODUCTS DEPARTMENT
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC

GEK–7608A (6–78) 1M(F)